

# Rishi Sangneria

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## Education

### University of Wisconsin-Madison

BS in Computer Science, BS in Mathematics

GPA: 3.56/4.0, Dean's List Fall 2024, Spring 2025

**Relevant Coursework:** Analysis I, Introduction to Algorithms, Probability Theory, Introduction to AI, Linear Algebra, Differential Equations, Multivariable Calculus, Discrete Mathematics

Madison, WI

May 2028

## Technical Skills

**Proficient Languages:** Python, Java, C, TypeScript, SQL

**Tools:** Git, AWS, PostgreSQL, Docker, Linux, React, [Next.js](#), pandas, NumPy, Matplotlib, scikit-learn, CHTC

## Technical Projects

### btcBillboard- BTC Pricing ML Model

May 2026 - Present

- Built ML trading system for **Kalshi BTC prediction markets** using **XGBoost** with isotonic calibration on 6 years of hourly OHLCV, on-chain, and sentiment data; 0.777 ROC-AUC; currently in paper-trading validation
- Engineered a **real-time pipeline** (data fetch → 25-feature vector → calibrated inference → signal → SQLite paper execution) running a live **5-min** loop with automated trade resolution
- Diagnosed and documented why **Black-Scholes** pricing baselines fail for short-horizon crypto markets; eliminated two model variants (sub-2h, 24h) after identifying noise-dominated signal and baseline-residual collapse.

### FireInTheHole- Custom Poker Variant EV Solver

May 2025 - May 2026

- Built a preflop 6-max 100BB **expected-value solver** for a custom poker variant: a **Monte Carlo equity engine**, frequency-weighted opponent range modeling, and fold/call/raise/jam EV computation across all 169 starting hands.
- Modeled opponent behavior with a pot-odds continuation function and designed range-narrowing and bluff-selection logic, sizing bluffs to a bet-size-dependent value-to-bluff ratio.
- Validated Monte Carlo equity against an exact-enumeration oracle to within 0.5% across an **84-test suite**

### Donna- Cross-Platform AI Assistant

December 2025-January 2026

- Built a cross-platform AI assistant using a **Tauri (Rust + TypeScript/React)** desktop client and **FastAPI (Python)** backend
- Enables natural-language creation of reminders and calendar events, inbox summarization, and email draft generation.
- Implemented secure **OAuth** integrations with Google and Microsoft for calendar and email access, keyboard-driven command invocation, and structured **REST endpoints** for task creation, event scheduling, and daily digest summaries.

## Work Experience

### Computational Biology and Machine Learning Lab

Madison, WI

#### Undergraduate Researcher

August 2025 – Present

- Working with Dr. Dhananjay Bhaskar (UW–Madison BME) and Dr. Samuel Brudner (Yale MCDB) on **Topological Data Analysis** of zebra-finch birdsong.
- Designed and trained a **self-supervised LSTM Autoencoder** to compute latent trajectories from raw zebra-finch birdsong audio across a dataset of **100+ recordings** using **Pytorch**.
- Performed recurrent-loop identification and pitch analysis on computed latent trajectories to characterize song structure and developmental patterns
- Presented research findings at the UW–Madison Undergraduate Research Symposium (April 2026).

## Leadership & Activities

### Phi Kappa Sigma International Fraternity

Madison, WI

#### Vice President

December 2025 – Present

- Oversee operations and day-to-day affairs for a **60+** member chapter and managing **~\$50K** in annual budget
- Supervising **15** subordinate officers in charge of operational, social, and philanthropic tasks related to the chapter.

### Humorology Inc.

Madison, WI

#### Director of "Zapkap" Subgroup

May 2025 – May 2026

- Directed a high-grade musical production with a **100+** member cast subgroup, coordinating rehearsals and logistics
- Led subgroup to raise **\$100K** in under 9 months; collectively raised **\$850K** for Gio's Garden Respite Care Center